

PROJECT REPORT

Flight Reservation System (FRS)

SUBMITTED TO

MD SAMSUDDOHA

Assistant Professor

Department of Computer Science and Engineering

University of Barishal

SUBMITTED BY

AMINUR RAHMAN

Roll: 05-003-03

Database - 03

Flight Reservation System (FRS)

A Flight Reservation System (FRS) is a digital platform that enables passengers to search, book, modify, and cancel flight tickets efficiently. Airlines and travel agencies use such systems to streamline booking processes, manage flight schedules, and ensure smooth passenger experiences. The system includes passengers, airlines, flights, airports, bookings and payments as its core entities.

1. Designing (Entity Relationship) ER Diagram

Step-1: Identify the Entities Required

For the Flight Reservation System (FRS), the core entities are:

1. **Passenger**
2. **Airline**
3. **Flight**
4. **Airport**
5. **Booking**
6. **Payment**

Step-2: Identify the Attributes and Primary key for each Entity

1. Passenger

- **Attributes:**
 - passenger_id (Primary Key)
 - name
 - gender
 - phone
 - email
 - dob (Date of Birth)
 - address
 - passport_number
- **Primary Key:** passenger_id

2. Airline

- **Attributes:**
 - airline_id (Primary Key)
 - name
 - contact_number
 - email
 - headquarters_address
- **Primary Key:** airline_id

3. Flight

- **Attributes:**

- flight_id (Primary Key)
- flight_number
- airline_id (Foreign Key referencing Airline)
- departure_airport_id (Foreign Key referencing Airport)
- arrival_airport_id (Foreign Key referencing Airport)
- departure_time
- arrival_time
- duration
- total_seats
- available_seats
- price

- **Primary Key:** flight_id

4. Airport

- **Attributes:**

- airport_id (Primary Key)
- name
- code (IATA code, e.g., JFK, LAX)
- city
- country
- contact_number

- **Primary Key:** airport_id

5. Booking

- **Attributes:**

- booking_id (Primary Key)
- passenger_id (Foreign Key referencing Passenger)
- flight_id (Foreign Key referencing Flight)
- booking_date
- seat_number
- booking_status (e.g., Confirmed, Cancelled, Pending)
- total_price

- **Primary Key:** booking_id

6. Payment

- **Attributes:**

- payment_id (Primary Key)
- booking_id (Foreign Key referencing Booking)
- payment_date
- payment_method (e.g., Credit Card, Debit Card, PayPal)
- amount
- payment_status (e.g., Paid, Pending, Failed)

- **Primary Key:** payment_id

Step-3: Identify the Relationships Needed

1. **Passenger-Books-Flight:** A passenger can book one or more flights.
2. **Flight-OperatedBy-Airline:** A flight is operated by one airline.
3. **Flight-DepartsFrom-Airport:** A flight departs from one airport.
4. **Flight-ArrivesAt-Airport:** A flight arrives at another airport.
5. **Booking-Has-Payment:** Each booking has a payment associated with it.

Step-4: Identify the Cardinality Ratio and Participation

1. Passenger-Books-Flight:



2. Flight-Operated By-Airline:



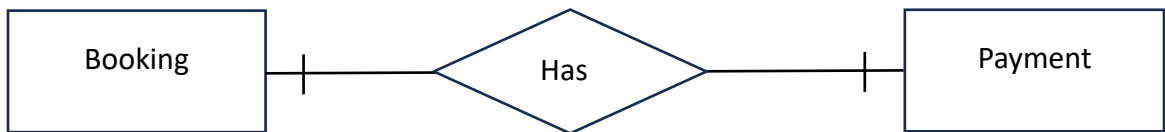
3. Flight-DepartsFrom-Airport:



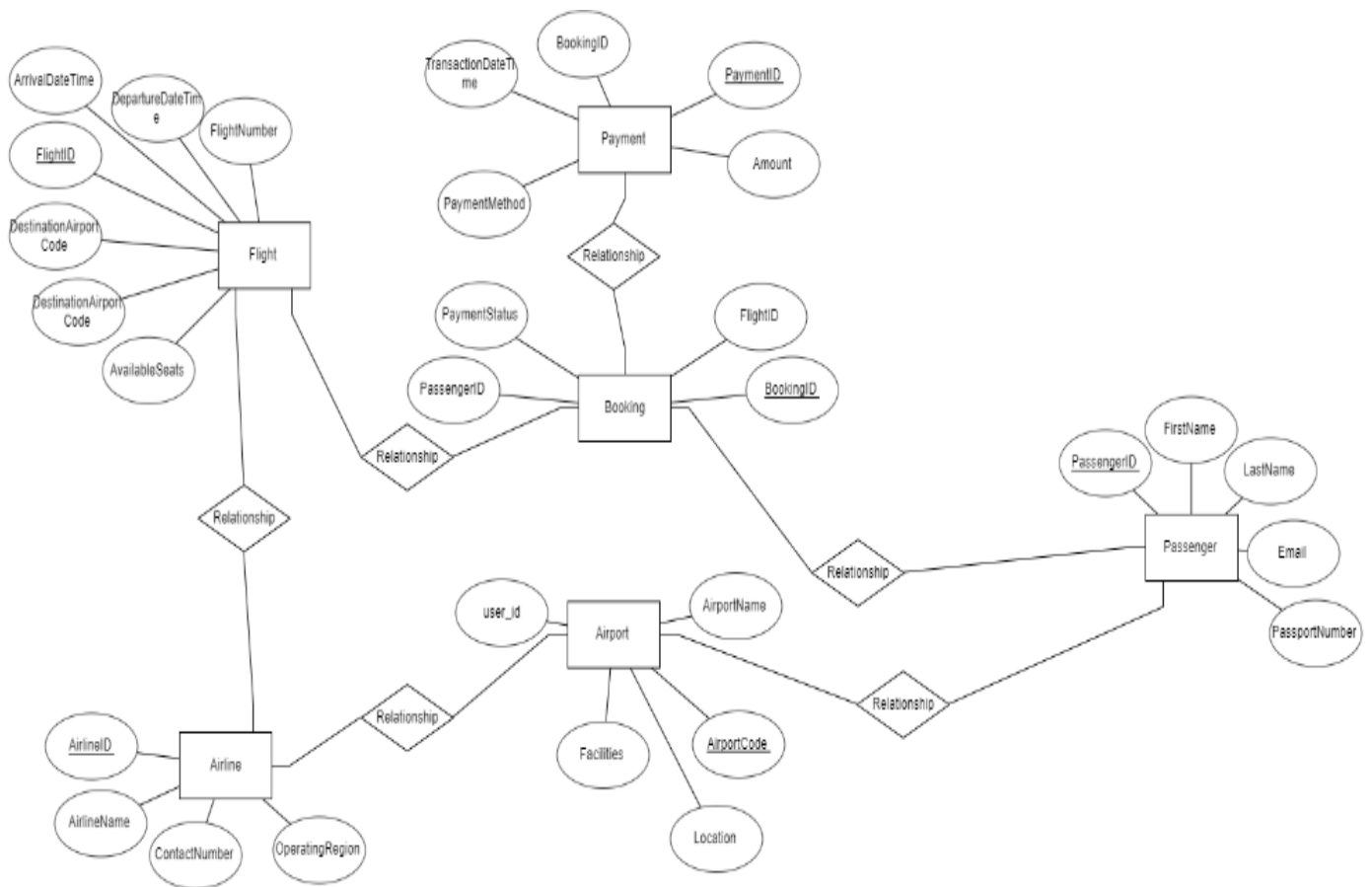
4. Flight-ArrivesAt-Airport:

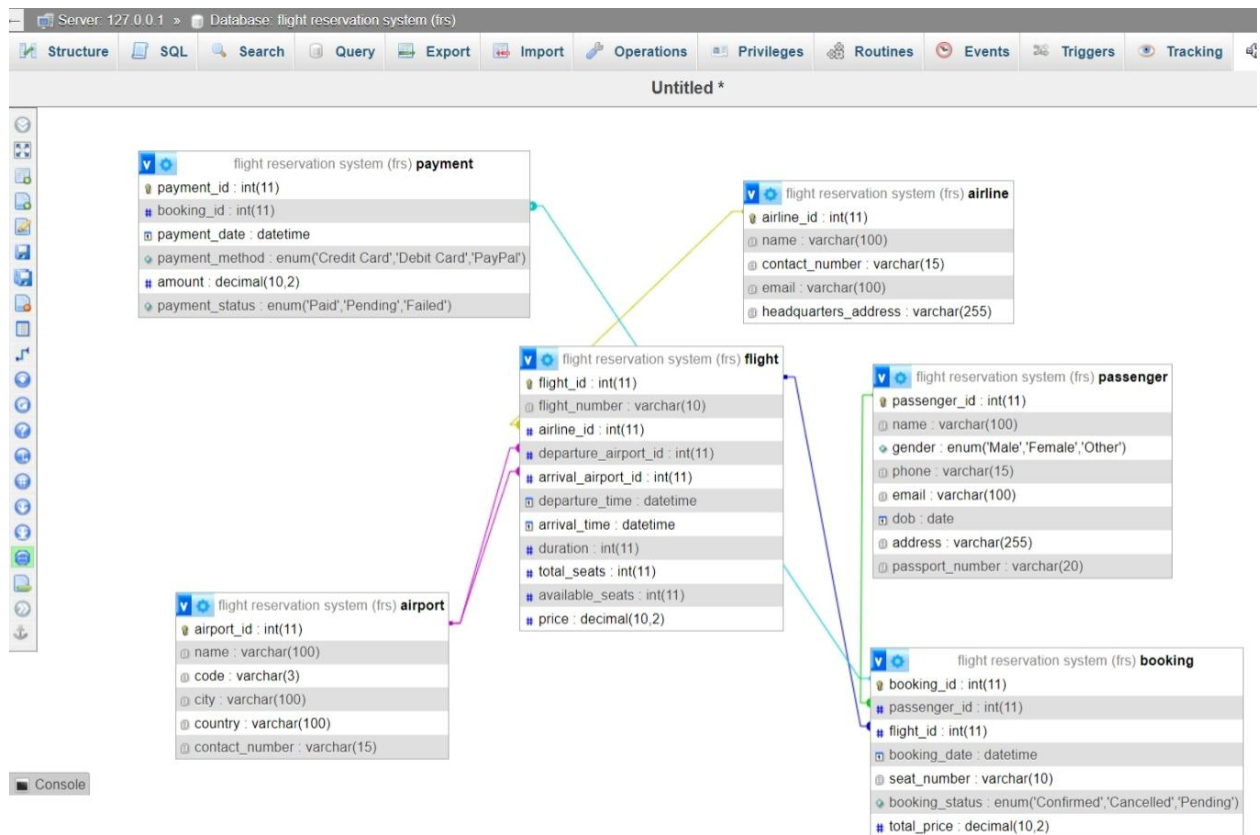


5. Booking-Has-Payment:



Step-5: Draw the ER Diagram





2. Reduction to Database Schema

Based on the entities and relationships, the database schema can be designed as follows:

- Passenger (`passenger_id`, name, gender, phone, email, dob, address, passport_number)
- Airline (`airline_id`, name, contact_number, email, headquarters_address)
- Flight (`flight_id`, flight_number, airline_id, departure_airport_id, arrival_airport_id, departure_time, arrival_time, duration, total_seats, available_seats, price)
- Airport (`airport_id`, name, code, city, country, contact_number)

- Booking (booking_id, passenger_id, flight_id, booking_date, seat_number, booking_status, total_price)
- Payment (payment_id, booking_id, payment_date, payment_method, amount, payment_status).

3.Implementing the database in MySQL:

All tables with sample data:

1. Passenger (passenger_id, name, gender, phone, email, dob, address, passport_number)

passenger_id	name	gender	phone	email	dob	address	passport_number
1	John Doe	Male	1234567890	john.doe@example.com	1990-01-01	123 Main St, City, Country	A12345678
2	Jane Smith	Female	0987654321	jane.smith@example.com	1995-05-15	456 Elm St, City, Country	B98765432
3	Alice Johnson	Female	1122334455	alice.johnson@example.com	1985-12-25	789 Oak St, City, Country	C11223344
4	Bob Brown	Male	5566778899	bob.brown@example.com	2000-07-30	321 Pine St, City, Country	D55667788

2. Airline (airline_id, name, contact_number, email, headquarters_address)

airline_id	name	contact_number	email	headquarters_address
1	Delta Airlines	1800123456	info@delta.com	Atlanta, Georgia, USA
2	American Airlines	1800654321	info@american.com	Fort Worth, Texas, USA
3	Emirates	1800999888	info@emirates.com	Dubai, UAE
4	British Airways	1800111222	info@britishairways.com	London, UK

3. Flight (flight_id, flight_number, airline_id, departure_airport_id, arrival_airport_id, departure_time, arrival_time, duration, total_seats, available_seats, price)

flight_id	flight_number	airline_id	departure_airport_id	arrival_airport_id	departure_time	arrival_time	duration	total_seats	available_seats	price
1	DL123	1	1	4	2023-10-01 10:00:00	2023-10-01 12:00:00	120	150	150	200.00
2	AA456	2	2	3	2023-10-02 14:00:00	2023-10-02 16:00:00	120	200	200	250.00
3	EK789	3	3	1	2023-10-03 18:00:00	2023-10-03 20:00:00	120	180	180	300.00
4	BA101	4	4	2	2023-10-04 22:00:00	2023-10-05 02:00:00	240	220	220	350.00

4. Airport (airport_id, name, code, city, country, contact_number)

airport_id	name	code	city	country	contact_number
1	John F. Kennedy International Airport	JFK	New York	USA	1800123456
2	Heathrow Airport	LHR	London	UK	1800111222
3	Dubai International Airport	DXB	Dubai	UAE	1800999888
4	Los Angeles International Airport	LAX	Los Angeles	USA	1800654321

5. Booking (booking_id, passenger_id, flight_id, booking_date, seat_number, booking_status, total_price)

booking_id	passenger_id	flight_id	booking_date	seat_number	booking_status	total_price
1	1	1	2023-09-25 14:00:00	A1	Confirmed	200.00
2	2	2	2023-09-26 15:00:00	B2	Confirmed	250.00
3	3	3	2023-09-27 16:00:00	C3	Pending	300.00
4	4	4	2023-09-28 17:00:00	D4	Cancelled	350.00

6. Payment (payment_id, booking_id, payment_date, payment_method, amount, payment_status).

payment_id	booking_id	payment_date	payment_method	amount	payment_status
1	1	2023-09-25 14:05:00	Credit Card	200.00	Paid
2	2	2023-09-26 15:05:00	Debit Card	250.00	Paid
3	3	2023-09-27 16:05:00	PayPal	300.00	Pending
4	4	2023-09-28 17:05:00	Credit Card	350.00	Failed